

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular/Supplementary Winter Examination – 2024**

**Course: B.Tech**

**Branch : Civil Engineering**

**Semester : III**

**Subject Code & Name: (BTCVES302) Mechanics of Solids**

**Max Marks: 60**

**Date: 07/02/2025**

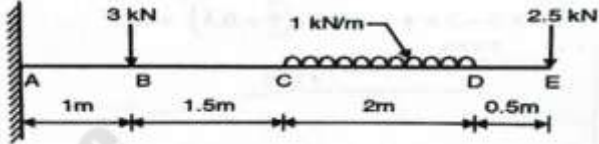
**Duration: 3 Hr.**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|                   |                                                                                                                                                      | (Level/CO)                  | Marks                      |                             |                  |  |  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|-----------------------------|------------------|--|--|
| <b>Q. 1</b>       | <b>Objective type questions. (Compulsory Question)</b>                                                                                               |                             | <b>12</b>                  |                             |                  |  |  |
| 1                 | The ratio of stress to strain in the linear region of a material's stress-strain curve is:                                                           | <b>L1/CO1</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. Shear modulus</td> <td>b. Young's modulus</td> <td>c. Poisson's ratio</td> <td>d. Bulk modulus</td> </tr> </table>    | a. Shear modulus            | b. Young's modulus         | c. Poisson's ratio          | d. Bulk modulus  |  |  |
| a. Shear modulus  | b. Young's modulus                                                                                                                                   | c. Poisson's ratio          | d. Bulk modulus            |                             |                  |  |  |
| 2                 | The unit of strain is:                                                                                                                               | <b>L1/CO1</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. Pascal</td> <td>b. Newton</td> <td>c. Joule</td> <td>d. Dimensionless</td> </tr> </table>                             | a. Pascal                   | b. Newton                  | c. Joule                    | d. Dimensionless |  |  |
| a. Pascal         | b. Newton                                                                                                                                            | c. Joule                    | d. Dimensionless           |                             |                  |  |  |
| 3                 | Which of the following materials would most likely have a high Young's modulus?                                                                      | <b>L1/CO1</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. Rubber</td> <td>b. Steel</td> <td>c. Wood</td> <td>d. Plastic</td> </tr> </table>                                     | a. Rubber                   | b. Steel                   | c. Wood                     | d. Plastic       |  |  |
| a. Rubber         | b. Steel                                                                                                                                             | c. Wood                     | d. Plastic                 |                             |                  |  |  |
| 4                 | Hooke's Law is valid only in the:                                                                                                                    | <b>L1/CO1</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. Plastic region</td> <td>b. Fracture region</td> <td>c. Elastic region</td> <td>d. Yield region</td> </tr> </table>    | a. Plastic region           | b. Fracture region         | c. Elastic region           | d. Yield region  |  |  |
| a. Plastic region | b. Fracture region                                                                                                                                   | c. Elastic region           | d. Yield region            |                             |                  |  |  |
| 5                 | A rod has a length of 2 meters and undergoes an elongation of 0.002 meters when subjected to a tensile load. What is the strain in the rod?          | <b>L2/CO1</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. 0.001</td> <td>b. 0.01</td> <td>c. 0.1</td> <td>d. 1</td> </tr> </table>                                              | a. 0.001                    | b. 0.01                    | c. 0.1                      | d. 1             |  |  |
| a. 0.001          | b. 0.01                                                                                                                                              | c. 0.1                      | d. 1                       |                             |                  |  |  |
| 6                 | For a cantilever beam subjected to a point load at the free end, the bending moment at the fixed support is:                                         | <b>L2/CO2</b>               | 1                          |                             |                  |  |  |
|                   | <table border="1"> <tr> <td>a. Zero</td> <td>b. Independent of the load</td> <td>c. Equal to the shear force</td> <td>d. Maximum</td> </tr> </table> | a. Zero                     | b. Independent of the load | c. Equal to the shear force | d. Maximum       |  |  |
| a. Zero           | b. Independent of the load                                                                                                                           | c. Equal to the shear force | d. Maximum                 |                             |                  |  |  |

|    |                                                                                         |                                                             |                                                        |                                          |        |   |
|----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------|------------------------------------------|--------|---|
| 7  | Which assumption is valid in the derivation of the torsion formula for circular shafts? |                                                             |                                                        |                                          | L2/CO4 | 1 |
|    | a. Shaft material is homogeneous and isotropic                                          | b. Plane sections before torsion remain plane after torsion | c. Shear stress is directly proportional to the radius | d. All of the A, B, and C                |        |   |
| 8  | The power transmitted by a shaft is proportional to:                                    |                                                             |                                                        |                                          | L2/CO4 | 1 |
|    | a. The applied torque only                                                              | b. The rotational speed only                                | c. The product of torque and rotational speed          | d. The square of the rotational speed    |        |   |
| 9  | Rankine's formula is used for:                                                          |                                                             |                                                        |                                          | L2/CO3 | 1 |
|    | a. Short columns only                                                                   | b. Long columns only                                        | c. Both short and long columns                         | d. Columns with one end fixed            |        |   |
| 10 | Theories of failure are used to predict failure in materials under:                     |                                                             |                                                        |                                          | L2/CO4 | 1 |
|    | a. Simple loading conditions                                                            | b. Complex, multi-axial loading conditions                  | c. Static loads only                                   | d. Dynamic loads only                    |        |   |
| 11 | When a shaft is subjected to torsion, the angle of twist ( $\theta$ ) is:               |                                                             |                                                        |                                          | L2/CO4 | 1 |
|    | a. Directly proportional to the length of the shaft                                     | b. Inversely proportional to the polar moment of inertia    | c. Inversely proportional to the modulus of rigidity   | d. All of the A, B, and C                |        |   |
| 12 | The axial force in a beam is defined as:                                                |                                                             |                                                        |                                          | L1/CO2 | 1 |
|    | a. The internal force perpendicular to the cross-section                                | b. The internal force parallel to the cross-section         | c. The internal torque on the beam section             | d. The sum of bending and shear stresses |        |   |
|    |                                                                                         |                                                             |                                                        |                                          |        |   |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Q. 2</b> | <b>Solve the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |               | <b>12</b> |
| <b>A)</b>   | Define stress, strain, and elasticity. Derive a relation between stress and strain of an elastic body.                                                                                                                                                                                                                                                                                                                                                  | <b>L2/CO1</b> | <b>6</b>  |
| <b>B)</b>   | A composite bar is made up of a steel section and a copper section, both of equal length (500 mm) and connected end-to-end. The steel section has a cross-sectional area of 200 mm <sup>2</sup> , and the copper section has a cross-sectional area of 250 mm <sup>2</sup> . A load of 30 kN is applied to the bar in tension. If the Young's modulus for steel is 200GPa and for copper is 120GPa, calculate the total extension of the composite bar. | <b>L3/CO1</b> | <b>6</b>  |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |               | <b>12</b> |
| <b>A)</b>   | Draw SFD and BMD for the beam shown in the figure:<br>                                                                                                                                                                                                                                                                                                                 | <b>L4/CO2</b> | <b>6</b>  |
| <b>B)</b>   | A simply supported beam of length 10m, carries two point loads of 15kN and 20kN at distance of 3m and 6m from left end. Draw SFD and BMD for beam.                                                                                                                                                                                                                                                                                                      | <b>L4/CO2</b> | <b>6</b>  |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                  |               | <b>12</b> |
| <b>A)</b>   | Derive the bending equation or flexural formula.                                                                                                                                                                                                                                                                                                                                                                                                        | <b>L2/CO4</b> | <b>6</b>  |
| <b>B)</b>   | A simply supported rectangular beam of width 100 mm, height 200 mm, and length 3 m is subjected to a uniform bending moment of 5 kNm. Calculate the maximum bending stress in the beam.                                                                                                                                                                                                                                                                 | <b>L3/CO4</b> | <b>6</b>  |
| <b>C)</b>   | A T-beam has the following dimensions: <ul style="list-style-type: none"> <li>Flange width = 150 mm</li> <li>Flange thickness = 20 mm</li> <li>Web height = 180 mm</li> <li>Web thickness = 10 mm</li> </ul> Calculate the section modulus of the T-beam about its horizontal neutral axis.                                                                                                                                                             | <b>L3/CO4</b> | <b>6</b>  |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                  |               | <b>12</b> |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                             |               |           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>A)</b>   | Explain Euler's column theory. Also write assumptions made in the theory.                                                                                                                                                                                                                                                                                                                                                   | <b>L2/CO3</b> | <b>6</b>  |
| <b>B)</b>   | An I-section joist of 500 mm * 250 mm * 20 mm and 5 m long is used as a strut. What is Euler's crippling load for the column? Take Young's modulus for the joist as 200 GPa                                                                                                                                                                                                                                                 | <b>L3/CO3</b> | <b>6</b>  |
| <b>C)</b>   | Compare the crippling loads given by Rankine's formula and Euler's formula for a tubular strut 3 m long having outer diameter 47.5 mm and 42.5 mm as inner diameter. Assume both ends pin-jointed. Take $\sigma = 315$ MPa and $\alpha = 1/7500$ .                                                                                                                                                                          | <b>L3/CO3</b> | <b>6</b>  |
| <b>Q. 6</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                      |               | <b>12</b> |
| <b>A)</b>   | Explain the following theories of failure, highlighting the criteria for failure:<br>(a) Maximum Principal Stress Theory<br>(b) Maximum Principal Strain Theory                                                                                                                                                                                                                                                             | <b>L2/CO4</b> | <b>6</b>  |
| <b>B)</b>   | Using Mohr's Circle, explain the graphical representation of principal stresses and strains. How can it be used to find the maximum shear stress?                                                                                                                                                                                                                                                                           | <b>L3/CO4</b> | <b>6</b>  |
| <b>C)</b>   | A machine component is subjected to a tensile stress of 300MPa in the horizontal direction and a compressive stress of 200MPa in the vertical direction. Additionally, a shear stress of 120MPa acts on the plane, tending to rotate the element in a clockwise direction. Determine the principal stresses and their directions, as well as the maximum shear stresses and the directions of the planes on which they act. | <b>L3/CO4</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                                                                                          |               |           |